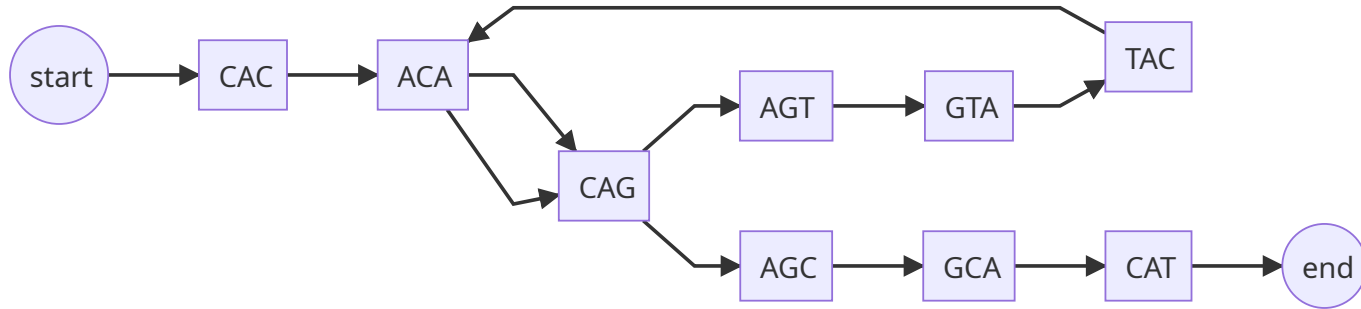


BIOL8701: Dividing and conquering sequence alignment using braided De Bruijn Graphs



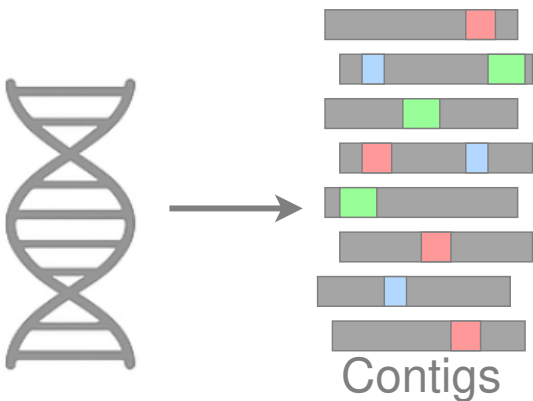
- Student: Richard Morris
- Huttley lab, Australian National University
- Supervisors: Gavin Huttley

Where does alignment fit in genomics?

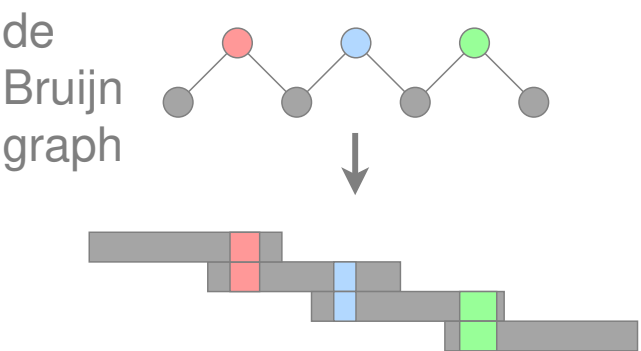
Sample



Amplification



Assembly



Alignment

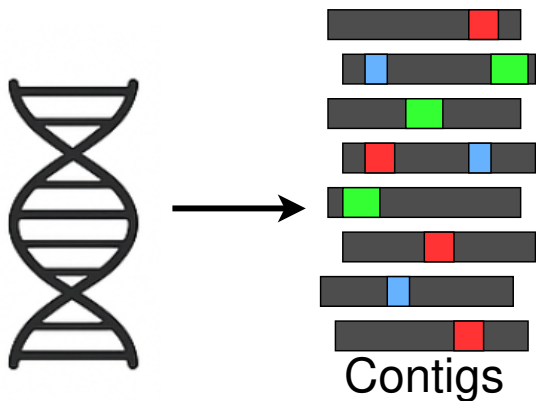


Where does alignment fit in genomics?

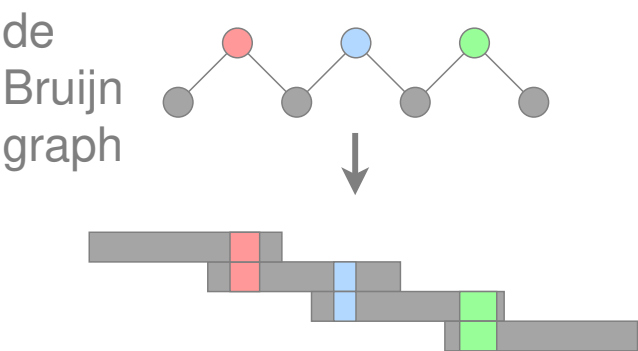
Sample



Amplification



Assembly



Alignment

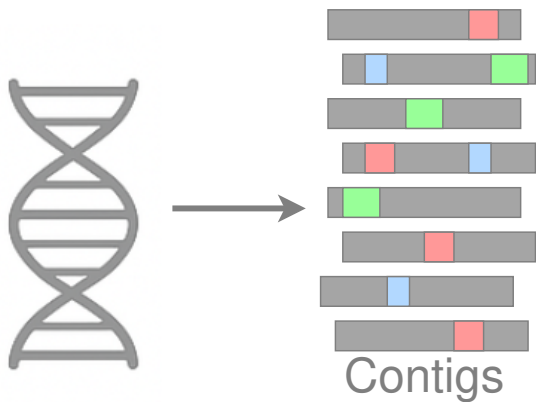


Where does alignment fit in genomics?

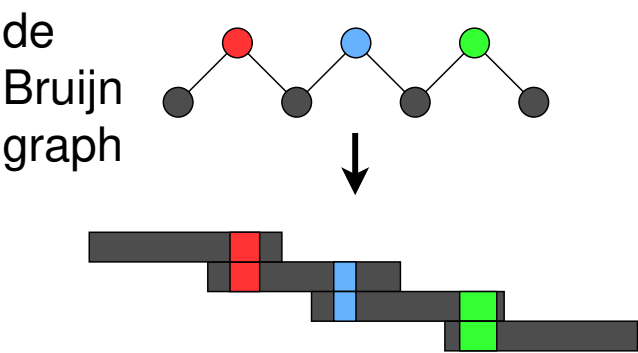
Sample



Amplification



Assembly



Alignment

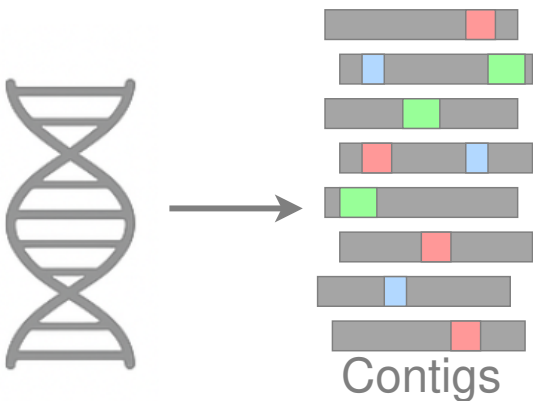


Where does alignment fit in genomics?

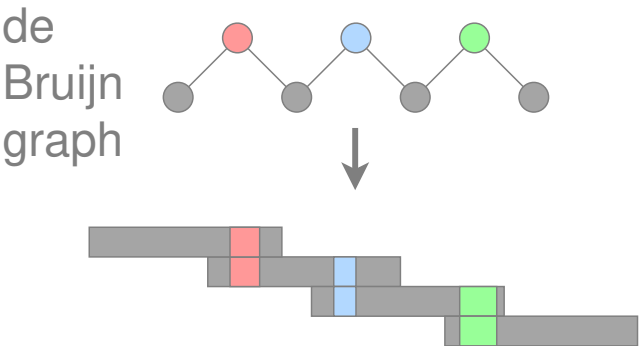
Sample



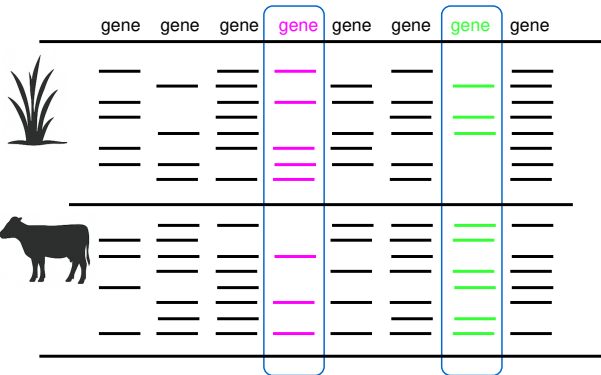
Amplification



Assembly



Alignment



Algorithm: Needleman-Wunsch¹ pairwise alignment

		G	A	T	T	A	C
G							
T							
T							
A							
G							
C							
C							

¹Needleman & Wunsch, 1970 [doi.org/10.1016/0022-2836\(70\)90057-4](https://doi.org/10.1016/0022-2836(70)90057-4)

Algorithm: Needleman-Wunsch pairwise alignment

		G	A	T	T	A	C
	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
T	-4	-1	0				
T							
A							
G							
C							
C							

Algorithm: Needleman-Wunsch pairwise alignment

		G	A	T	T	A	C
	0	-2	-4	-6	-8	-10	-12
G	-2	1	-1	-3	-5	-7	-9
T	-4	-1	0	0	-2	-4	-6
T	-6	-3	-2	1	1	-1	-3
A	-8	-5	-2	-1	0	2	0
G	-10	-7	-4	-3	-2	0	1
C	-12	-9	-6	-5	-4	-2	1
C	-14	-11	-8	-7	-6	-4	-1

Alignment: global
Performance: $O(L^2)$

Result:

G	-	T	T	A	G	C	C
*		*	*	*			*
G	A	T	T	A	-	-	C

Algorithm: Smith Waterman² local alignment

		G	A	T	T	A	C
		0	0	0	0	0	0
G		0	1	0	0	0	0
T		0	0	0	1	1	0
T		0	0	0	1	2	0
A		0	0	1	0	0	3
G		0	1	0	0	0	1
C		0	0	0	0	0	2
C		0	0	0	0	0	1

Alignment: local

Performance: $O(L^2)$

Result:

T	T	A
*	*	*
T	T	A

²Smith & Waterman, 1981 10.1016/0022-2836(81)90087-5

Multiple sequence alignment

“Multiple sequence alignment is just pairwise alignment, repeated many times”.

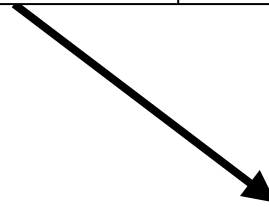
Progressive multisequence alignment: $O(nL^2)$

- L = length of sequence
- n = number of sequences

Algorithm: de Bruijn³ graphs

Sequence

B	I	O	L	O	G	Y
----------	----------	----------	----------	----------	----------	----------



Performance: $O(L)$

de Bruijn graph ($k = 3$)



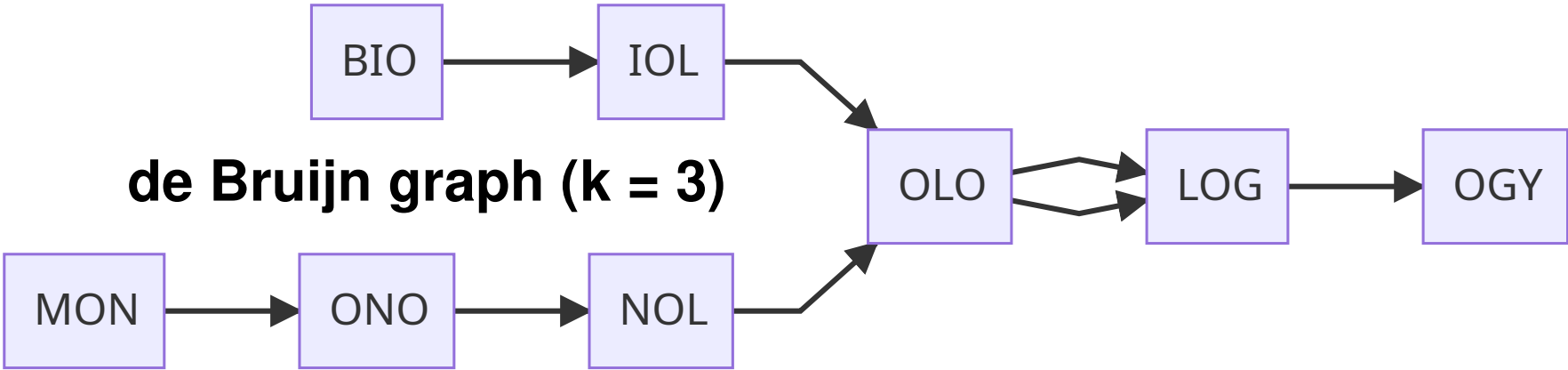
Algorithm: de Bruijn graphs

Sequence 1

B	I	O	L	O	G	Y
---	---	---	---	---	---	---



de Bruijn graph ($k = 3$)



Sequence 2

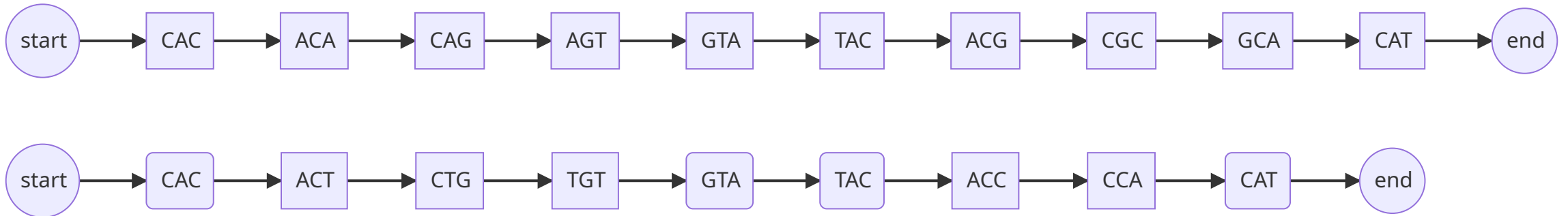
M	O	N	O	L	O	G
---	---	---	---	---	---	---



Algorithm: de Bruijn graphs

Sequence A: CACAGTACGCAT

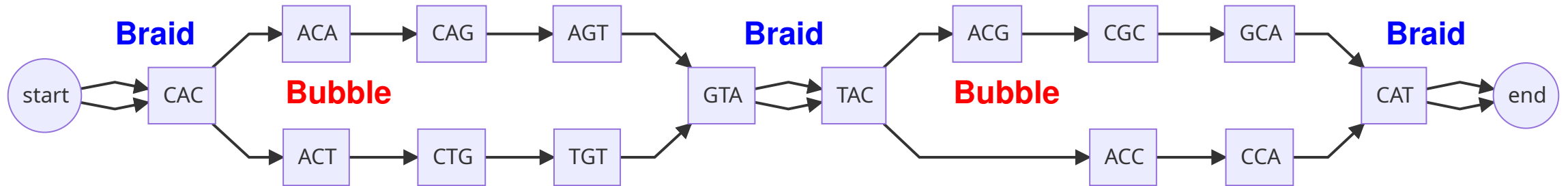
Sequence B: CACTGTACCAT



Algorithm: de Bruijn graphs

Sequence A: CACAGTACGCAT

Sequence B: CACTGTACCAT



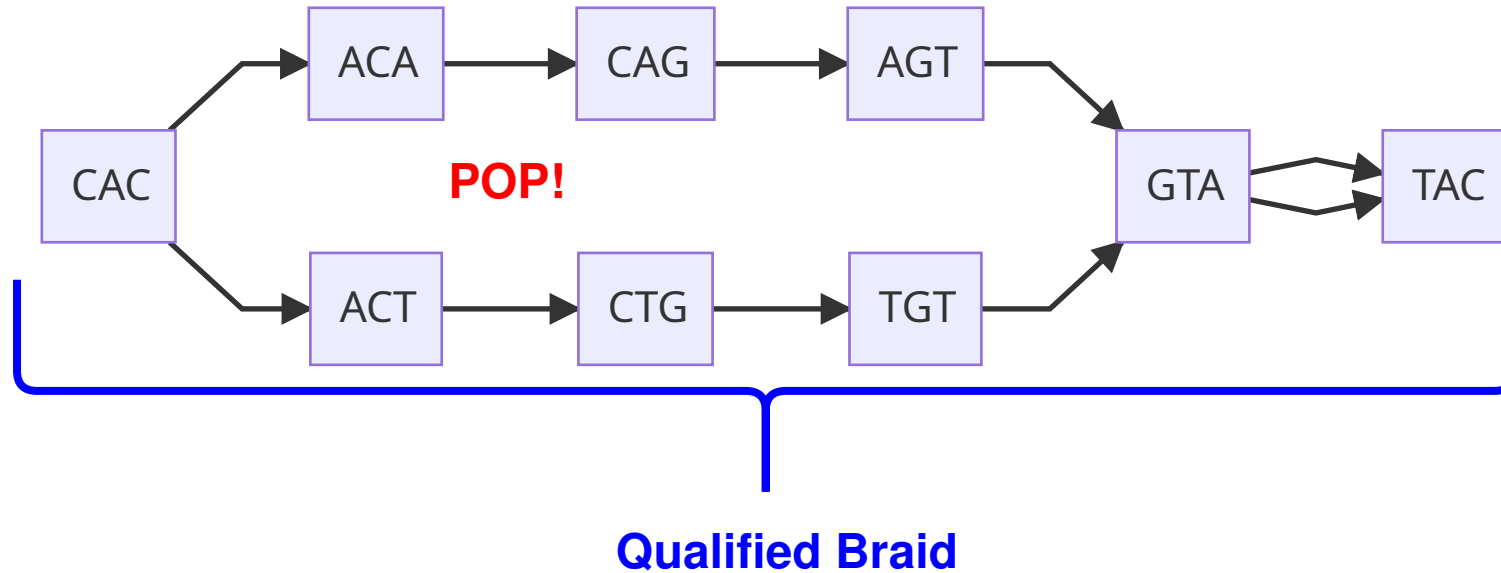
3 Braids: Already aligned

2 Bubbles: need to be aligned

Algorithm: de Bruijn graphs

Sequence A: CACAGTACGCAT

Sequence B: CACTGTACCAT



Algorithm: Karlin Altschul⁴ test

The Karlin–Altschul test is used to assess the significance of local sequence alignments.

Models the distribution of maximum alignment scores between two random sequences using a Gumbel (extreme value) distribution

$$E = K.m.n.e^{-\lambda S}$$

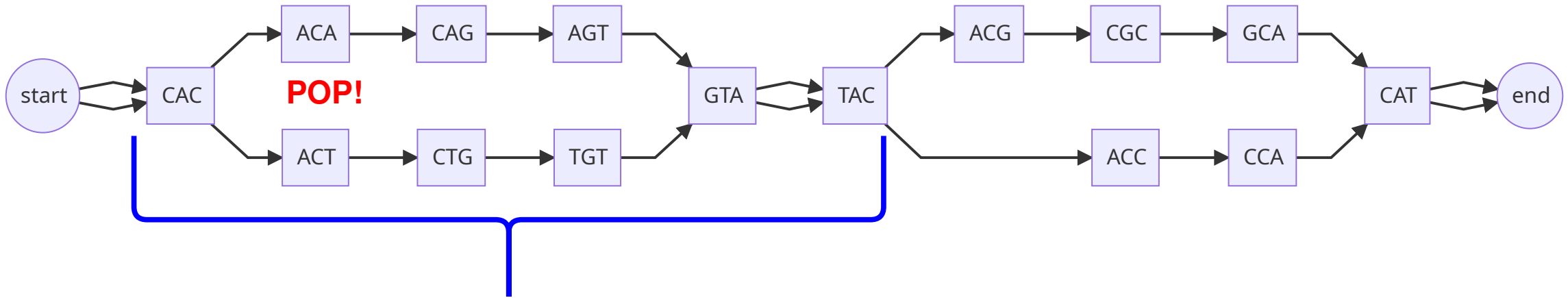
- E : expected number of alignments with score $\geq S$
- K : scale & λ : decay (dependent on scoring scheme)
- S : alignment score
- m : length of sequence 1 & n : length of sequence 2

⁴Karlin & Altschul, 1990 doi.org/10.1073/pnas.87.6.2264

Algorithm: Karlin Altschul test for biologists

Whether a local alignment, containing substitutions, meets a threshold (p-value) for equality.

ie: If the local alignment `CAC|A|GTAC` & `CAC|T|GTAC` meets the threshold, it is a **qualified braid**.



Qualified Braid

Karlin-Altschul ₁ test

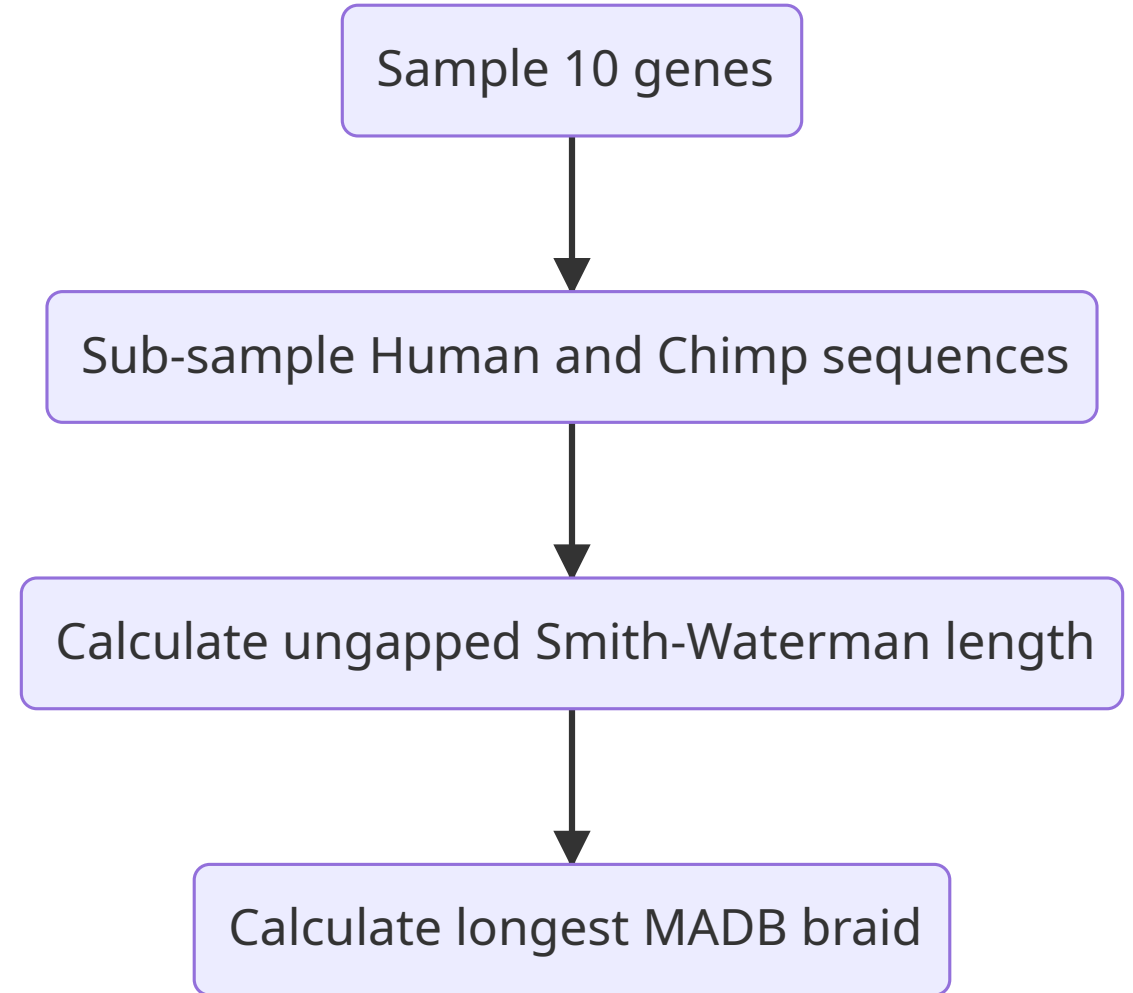
Base data and methods

- Mammal genomes obtained from **Ensembl**: release 113 (current release)
- **Queried**: using `ensembl-tui` application
- **Extracted**: 1597 one-to-one orthologs to human chromosome 1 protein coding genes
- **Alignment**
 - cogent3 (Smith-Waterman, Needleman-Wunsch)
 - madb (de Bruijn graph)

Hypothesis 1

The longest braid in a pair of sequences corresponds to the ungapped Smith-Waterman local alignment.*

Experimental design:



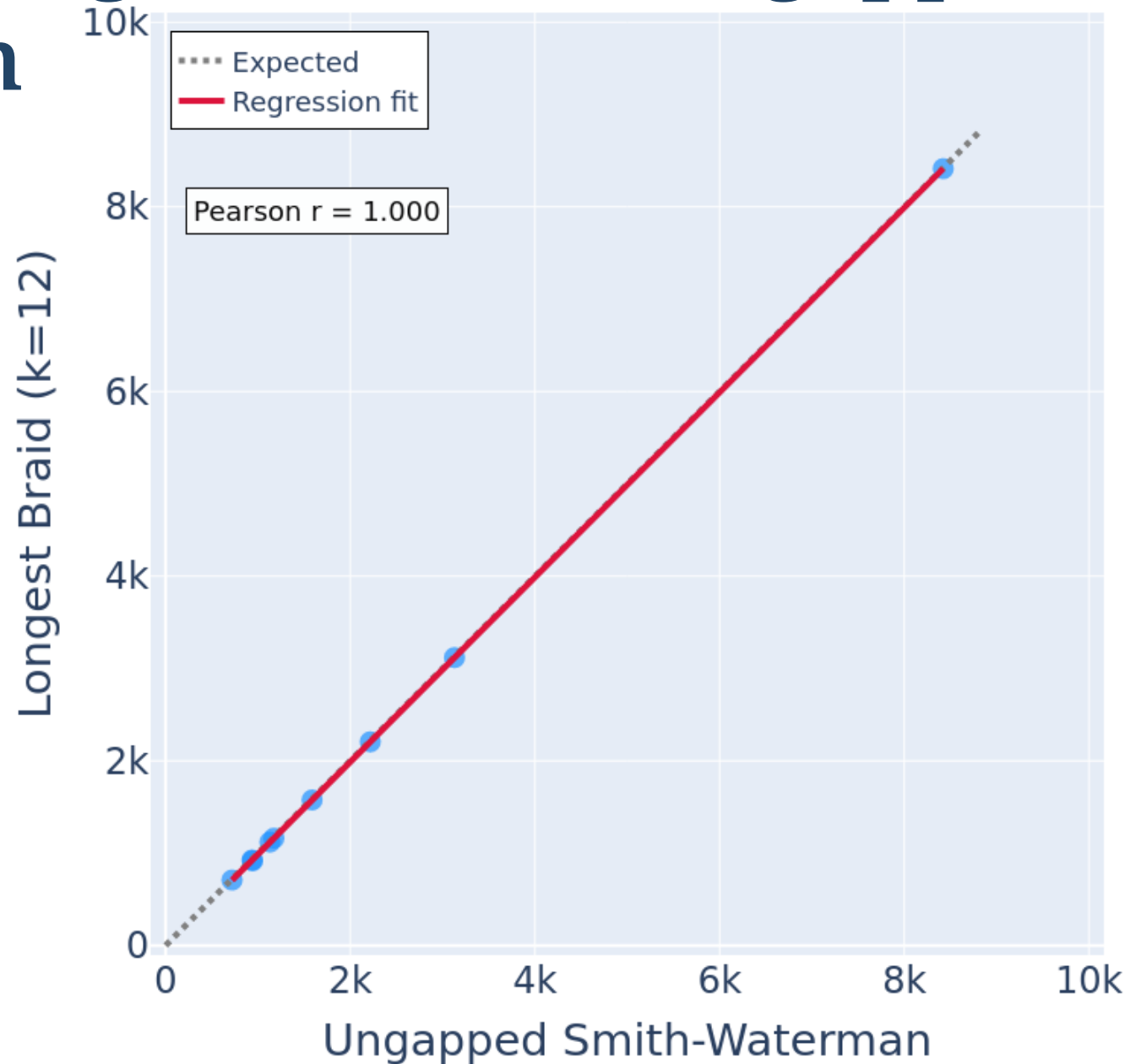
* a gap not within k characters of the start or end of a bubble ...

Background for Hypothesis 1

A braid has no gaps, this corresponds to an ungapped local alignment

- the Smith-Waterman algorithm is an optimal solution for finding the highest scoring local alignment
- the longest braid in a de Bruijn graph should equal the Smith-Waterman alignment

Results - longest braid vs ungapped Smith-Waterman



Summary for hypothesis 1

Smith-Waterman

produces **just one** best local
alignment in quadratic time
 $O(L^2)$

de Bruijn graph

produces **every** local alignment
in linear time $O(L)$

Summary for hypothesis 1

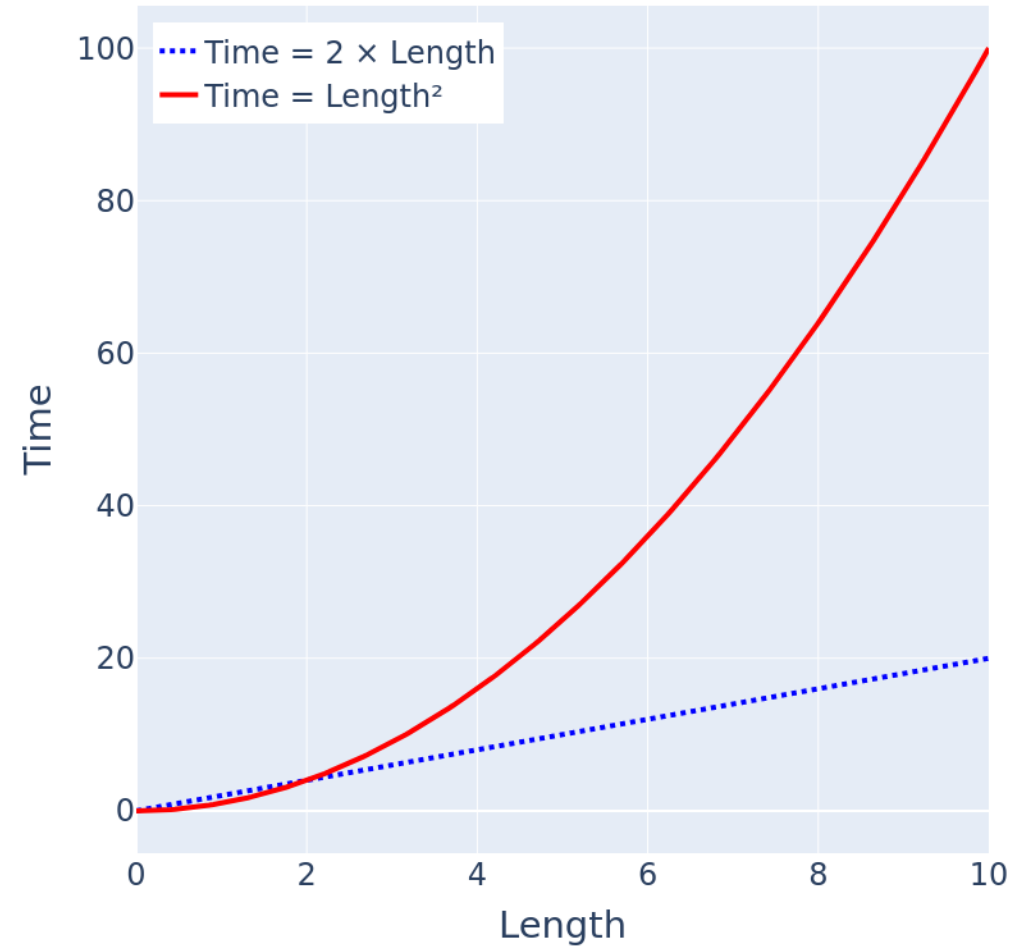
Smith-Waterman

produces **just one** best local alignment in quadratic time $O(L^2)$

Bruijin graph

produces **every** local alignment in linear time $O(L)$

Linear vs Quadratic Time Growth



Hypothesis 2

Differences in braids, will be a better approximation to the actual *genetic distance* than *Jaccard*.

Experimental design:

align Human Chimp sequences using cogent3

Calculate proportion different

degap and calculate jaccard distance

project to de Bruijn graph

count differences in braids/braid length

Background for Hypothesis 2

- Genetic distance is proportional to the time since two sequences diverged from a common ancestor
- Influences the probability that two nucleotides are a “match” (descended from a common ancestor)
- Required to construct reliable scoring functions for alignment methods

Term: *Genetic distance*

Consider the proportion different (PD) between 2 *aligned* sequences.

$$PD = \frac{\text{substitutions}}{\text{positions}}$$

0	
Human	T G C C A G C T C A T T A C A G C A T G A G A A C A G C A G T T T A T T A C T C A C T A A A G A C A
Mouse	 C G T . G
Wombat	 T T T . C . A . A . C G . C . . A C . C

3 x 50 dna alignment

$$PD_{human,mouse} = \frac{5}{50} = 0.1$$

$$PD_{human,wombat} = \frac{13}{50} = 0.26$$

Term: *Jaccard distance*

- Jaccard distance is a measure of dissimilarity between two sets, defined as the size of the intersection divided by the size of the union of the sets.
- Jaccard can be calculated for *unaligned* sequences, by decomposing them into k -mers and comparing the ratio of common k -mers to total k -mers

$$Jaccard = \frac{I}{U}$$

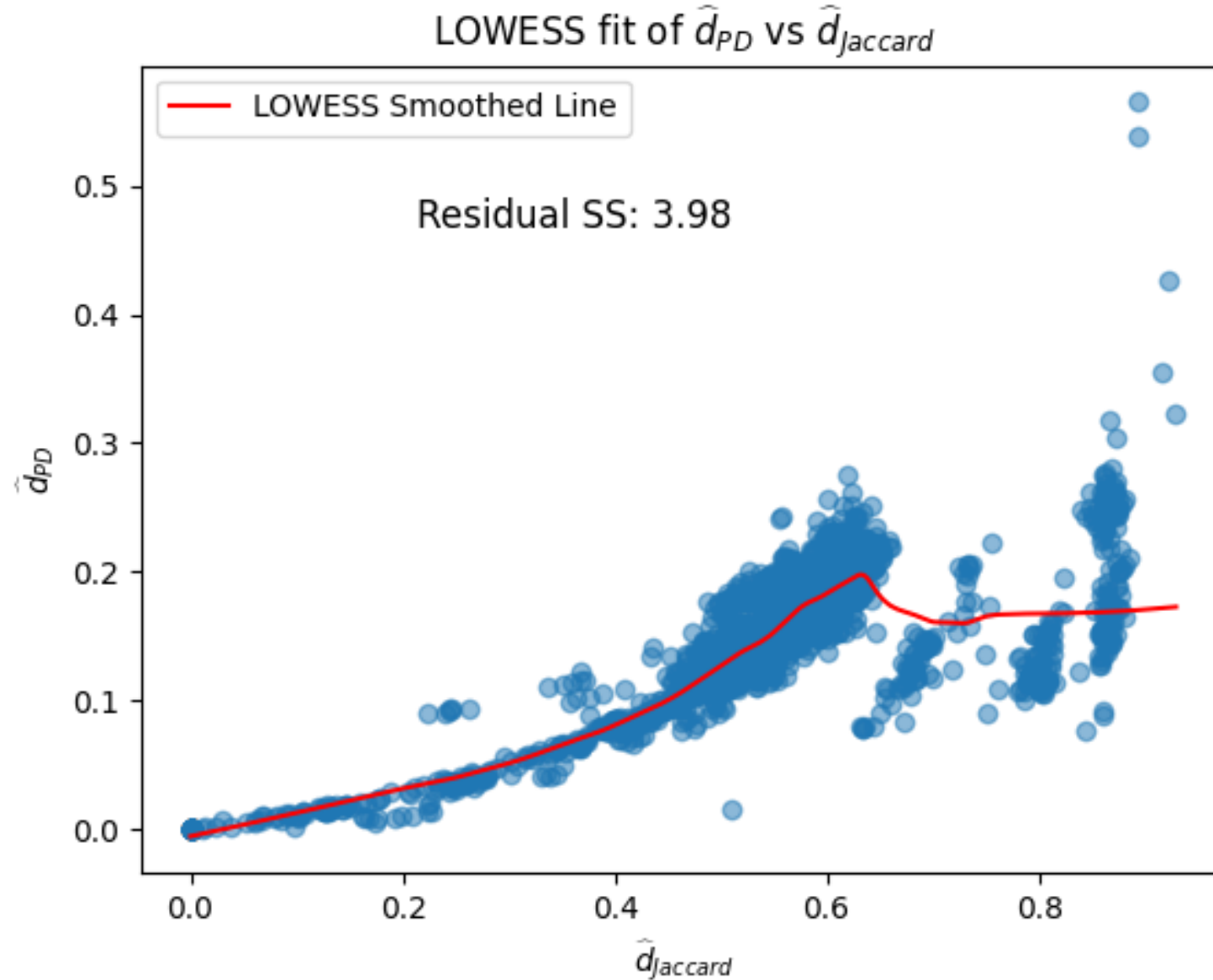
Key insight

By qualifying braid sequences per the *Karlin-Altshul* test, we are able to observe the proportion of nucleotides different in local alignments.

This offers a measure of genetic distance between “unaligned” sequences.

Thanks to Eric Stone for this valuable insight

Result 2 Jaccard distance vs PD



Term: *Stone metric (SM)*

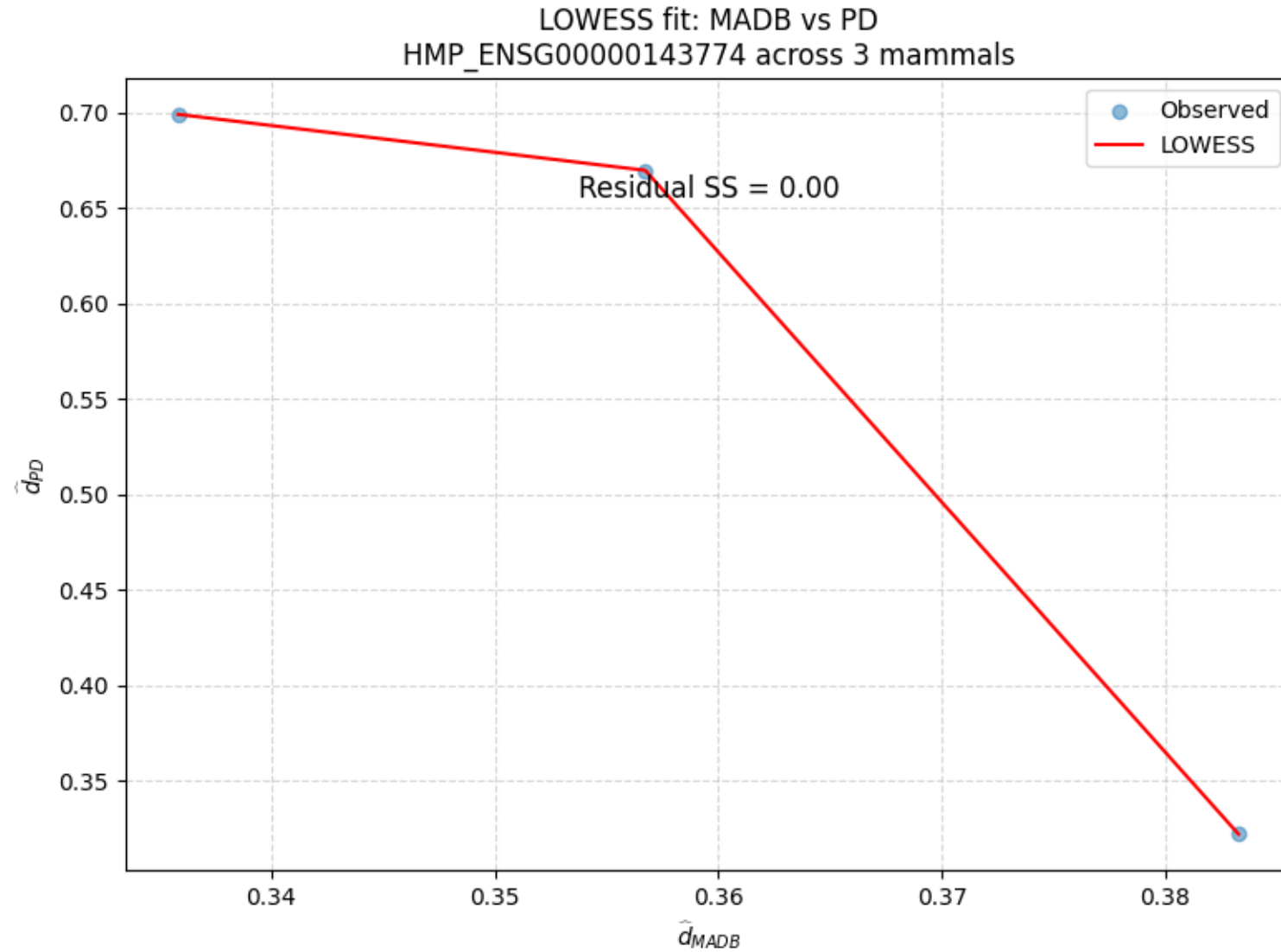
	0
Human	T G C C A G C T C A T T A C A G C A T G A G A A C A G C A G T T T A T T A C T C A C T A A A G A C A
Mouse	 C C G T . G
Wombat	 T T T . C . A . A . C G . C . . A C . C

3 x 50 dna alignment

The PD of all local alignments of the sequences will approach the PD of fully aligned sequences.

$$SM = \frac{\text{sum of Braid differences}}{\text{sum of Braid lengths}}$$

Result 2 Stone braid metric



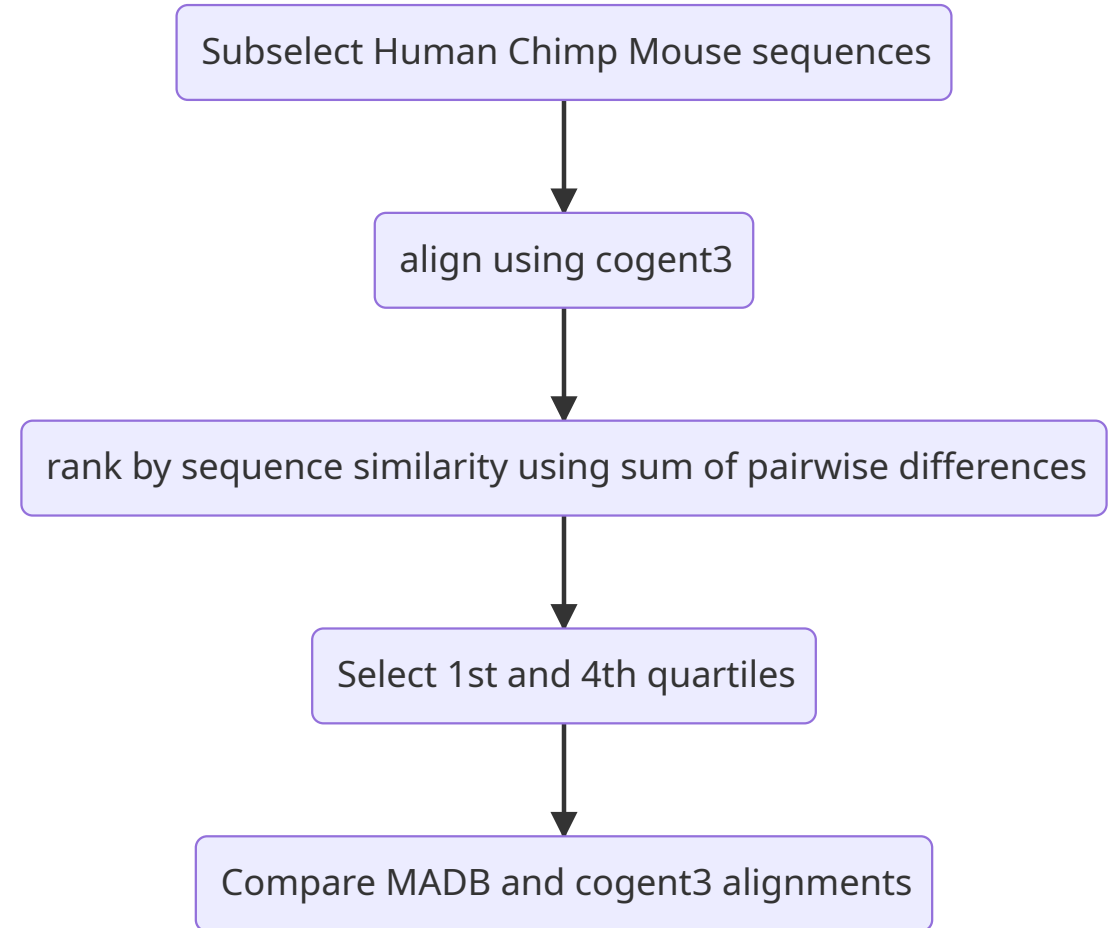
Summary for hypothesis 2

- Jaccard distance is a poor approximation of PD as it is noisy, and context dependent on the sequence being aligned
- The Stone metric has been affected by a bug we have in resolving cycles

Hypothesis 3

We predict that the de Bruijn graph aligner will perform (statistically and computationally) better with less diverged sequences.

Experimental design:



Background for Hypothesis 3

↑genetic distance $\Rightarrow$ ↑bubbles $\Rightarrow$ ↓braids $\Rightarrow$ ↑work

	0
Human	G T T T C A A A G C G C C A G T C A T T T G C T C C G T T T T C A A A T C C A G G A A A T G C A G A
Mouse	 T T T A A . . T A . . . G . C . C C .
Chimpanzee	. T C

3 x 50 dna alignment

Key insights:

- for a given sequence the number of cycles will decrease as you increase k
- our ability to find braids will improve as we decrease k

Result - TBD

MADB accuracy and performance by divergence

Human to Chimpanzee orthologous gene alignment

quartile	mean alignment score (%)	mean performance (s)
1st	TBD	TBD
4th	TBD	TBD

Human to Mouse orthologous gene alignment

quartile	mean alignment score (%)	mean performance (s)
1st	TBD	TBD
4th	TBD	TBD

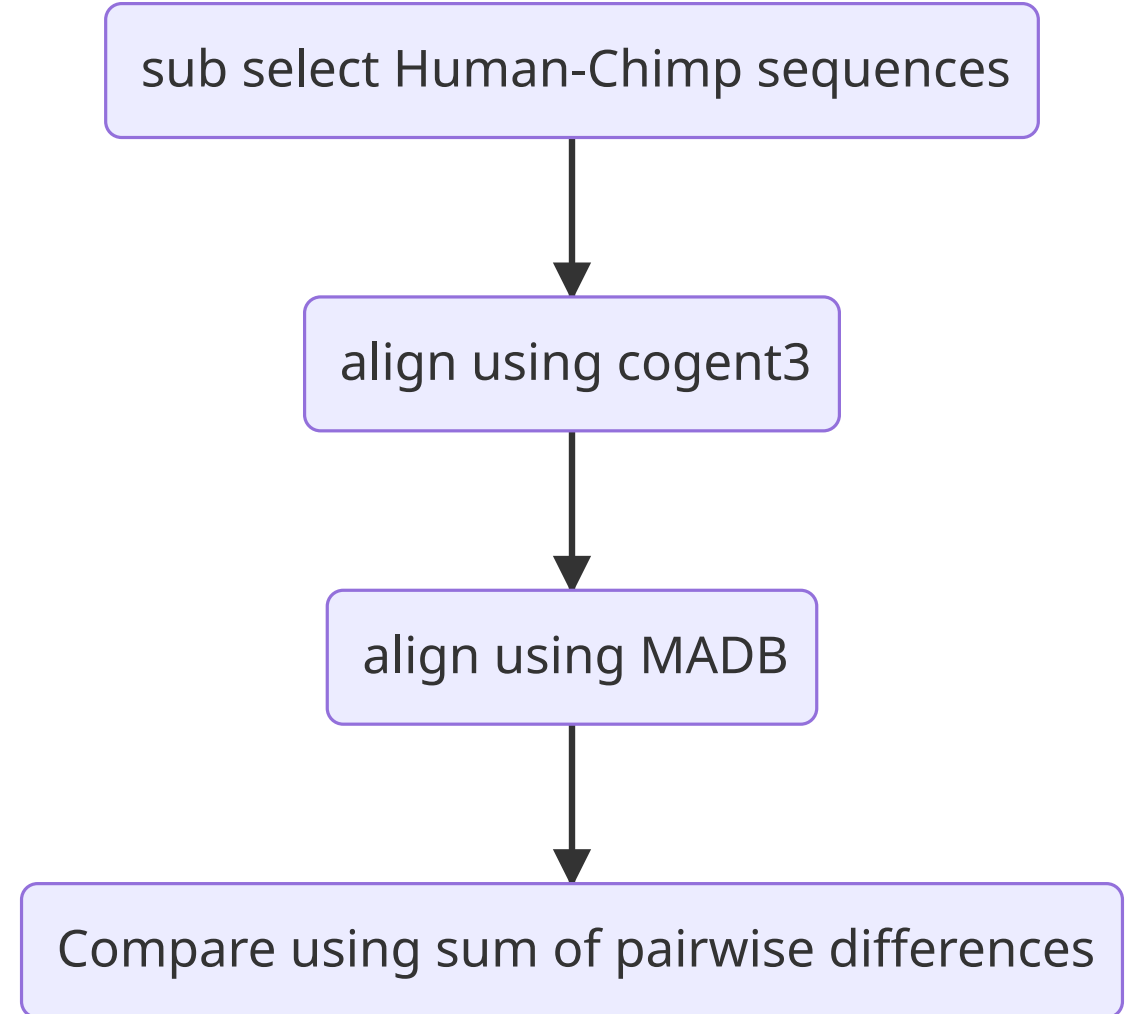
Summary for hypothesis 3

- the algorithm we are using to resolve cycles needs to be made more robust to sequence divergence
- next step is sampling another primate lineage more diverged from humans than chimpanzee such as marmoset

Hypothesis 4

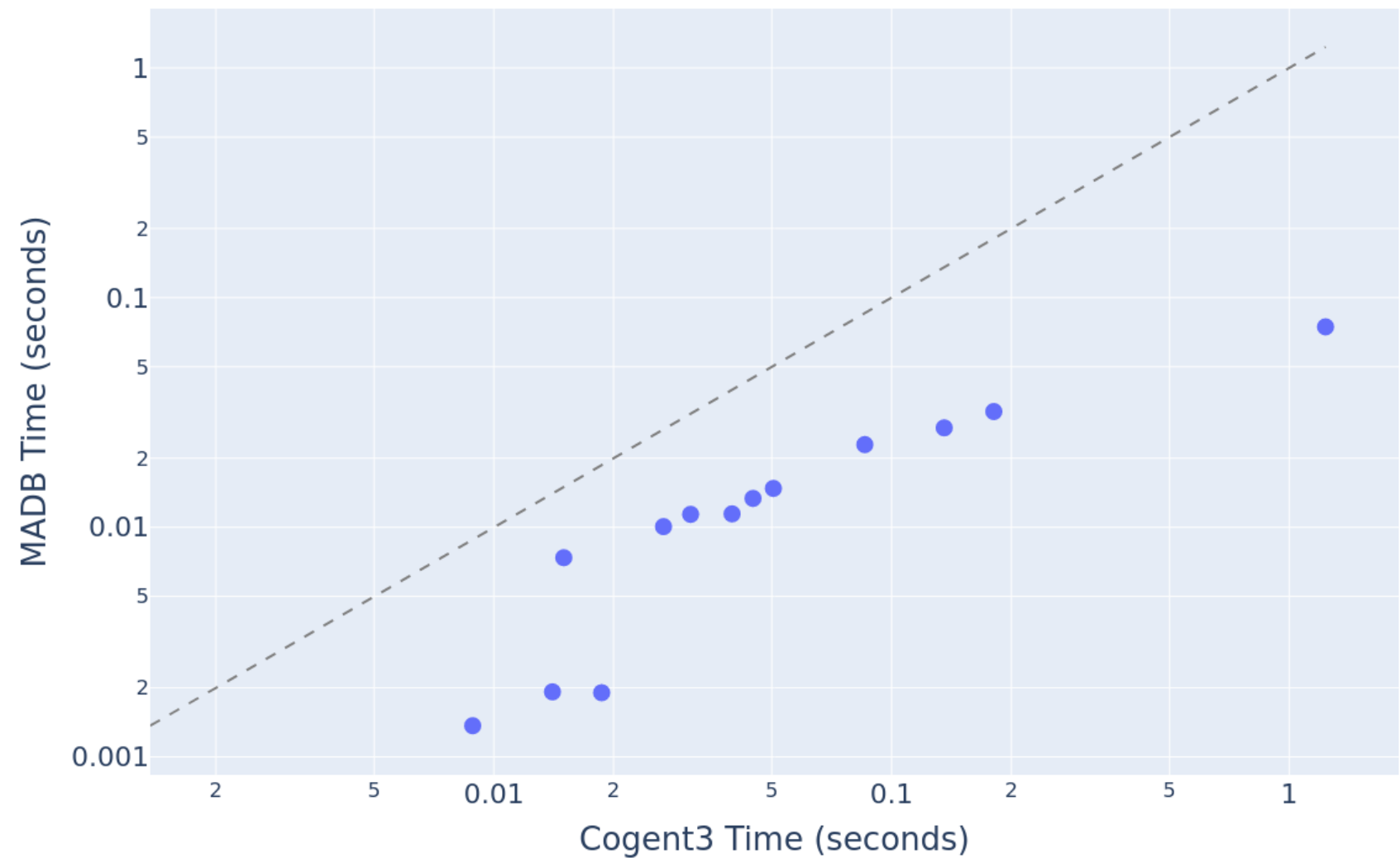
de Bruijn graph alignment will be more computationally efficient than Needleman-Wunsch.

Experimental design:



Results

Cogent3 vs MADB Alignment Time (log-log scale)



Summary for hypothesis 4

- de Bruijn graph alignment is more computationally efficient than Needleman-Wunsch because it reduces costly computation

Thanks

- Gavin Huttley
- Yu Lin
- Xinjian Leng

... and the Huttleylab



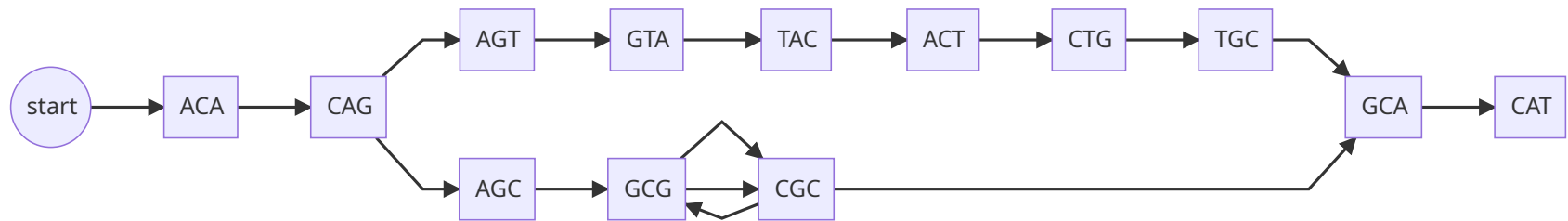
Questions & Answers

Citations

1. Needleman & Wunsch(1970), 'A general method applicable to the search for similarities in the amino acid sequence of two proteins' doi.org/10.1016/0022-2836(70)90057-4, 2010
2. Smith & Waterman(1981), 'Identification of Common Molecular Subsequences.' doi.org/10.1016/0022-2836(81)90087-5
3. de Bruijn(1946) 'A Combinatorial Problem.' Koninklijke Nederlandse Akademie van Wetenschappen, Proceedings 49 (1946): 758–64.
4. Karlin & Altschul(1990), 'Methods for assessing the statistical significance of molecular sequence features by using general scoring schemes' doi.org/10.1073/pnas.87.6.2264

Addendum - Removing cycles

Cyclic de Bruijn graph



Acyclic de Bruijn graphs

